A Tiny Robot That Swims With Living Muscle

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Scientists at MIT have built a tiny robot that swims with real, living muscle instead of batteries and motors.

The robot's body is soft and **flexible**, about the size of a stick of gum. It has two fins, and each fin is covered with **muscle cells** much thinner than a human hair. Scientists changed the cells so they squeeze tight when light shines on them. By flashing light on one fin or the other, the team can **steer** the robot and change its speed.

In tests, the robot swam and turned through a simple underwater **maze**. At its fastest, it moved about four body lengths every minute. An Olympic swimmer can move up to 65 body lengths a minute.

Study author Ritu Raman said it takes a lot of force to move through water, and the robot is quite strong for its size. The study appeared in the journal Advanced Functional Materials. One day, robots like this might do gentle jobs in **fragile** places, because living **tissue** is soft and can even heal itself.

Word Treasure

muscle cells -- Tiny living parts of muscle that can squeeze and pull to make things move.

flexible -- Able to bend easily without breaking.

steer -- To guide something in the direction you want it to go.

maze -- A puzzle of paths that you must find the right way through.

tissue -- A group of similar living cells that work together in a body.

fragile -- Easily broken or damaged, so it must be handled gently.

Background you need

Most robots move with motors and batteries, but some scientists are building a new kind of robot that uses living parts, like real muscle.

Muscle works by squeezing tight and then relaxing, and that squeezing can push a robot through the water.

Scientists like these soft, living robots because tissue is gentle and can even heal itself after it is hurt.

Break it down

WHO: Scientists at MIT, including study author Ritu Raman

WHAT: The team built a gum-sized robot with two fins covered in muscle cells that squeeze tight when light shines on them

WHERE: At MIT, with tests in a simple underwater maze

WHY: Because living tissue is soft and can heal itself, robots like this might one day do gentle jobs in fragile places

Why it matters

One day, soft robots like this could explore fragile places you care about, like coral reefs, without damaging them.

Test what you remember

1. What does the tiny robot use to swim instead of batteries and motors?

(A) Real, living muscle
(B) Tiny paper wings
(C) Magnets in its fins
(D) A small propeller

2. About how big is the robot?

(A) As big as a shoe
(B) About the size of a stick of gum
(C) Smaller than a grain of sand
(D) As big as a bicycle

3. How do scientists steer the robot?

(A) By flashing light on one fin or the other
(B) By pulling it on a string
(C) By blowing bubbles at it
(D) With a remote control joystick

4. What happens to the muscle cells when light shines on them?

(A) They change color
(B) They grow bigger
(C) They squeeze tight
(D) They fall asleep

5. In tests, what did the robot swim and turn through?

(A) A deep ocean trench
(B) A river full of rocks
(C) An Olympic swimming pool
(D) A simple underwater maze

6. Why might robots like this be good for gentle jobs in fragile places?

(A) Living tissue is soft and can heal itself
(B) They can swim faster than any fish
(C) They are made of hard metal
(D) They never need food or rest

Answer key (try the quiz first!): 1: A 2: B 3: A 4: C 5: D 6: A

Questions to think about

1. Study author Ritu Raman: She says it takes a lot of force to move through water, and the robot is quite strong for its size.

MY THOUGHTS (at least 20 words):